Lecture 13: Section 3.6

L'Hôpital's Rule

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What is L'Hôpital's Rule?

Definition

L'Hôpital's Rule is a method for evaluating limits of indeterminate forms by using derivatives.

The Big Idea

When direct substitution gives an indeterminate form, we can often find the limit by taking derivatives of the numerator and denominator.

Example

$$\lim_{x \to 0} \frac{\sin x}{x} = \lim_{x \to 0} \frac{\cos x}{1} = 1$$

Historical Note

Named after French mathematician Guillaume François Antoine de L'Hôpital (1661–1704), though it was actually discovered by Johann Bernoulli.

Indeterminate Forms

Forms Where L'Hôpital's Rule Applies

- $\frac{0}{0}$ (Most common)
- $\frac{\infty}{\infty}$
- $0 \cdot \infty$
- $\infty - \infty$
- 0^0, 1^∞, ∞^0

Forms That Are NOT Indeterminate

- $\frac{0}{\infty} = 0$
- $\frac{\infty}{0} = \infty$
- $0^\infty = 0$
- $\infty^\infty = \infty$

L'Hôpital's Rule Theorem

Suppose f and g are differentiable on an interval containing a (except possibly at a), and $g'(x) \neq 0$ near a. If:

$$\lim_{x \to a} \frac{f(x)}{g(x)} \text{ is of the form } \frac{0}{0} \text{ or } \frac{\infty}{\infty}$$

then:

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$$

provided the limit on the right exists or is $\pm\infty$.

Important Conditions

- Must have $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form
- Functions must be differentiable near a
- $g'(x) \neq 0$ near a

Type 1: $\frac{0}{0}$ Form

Example

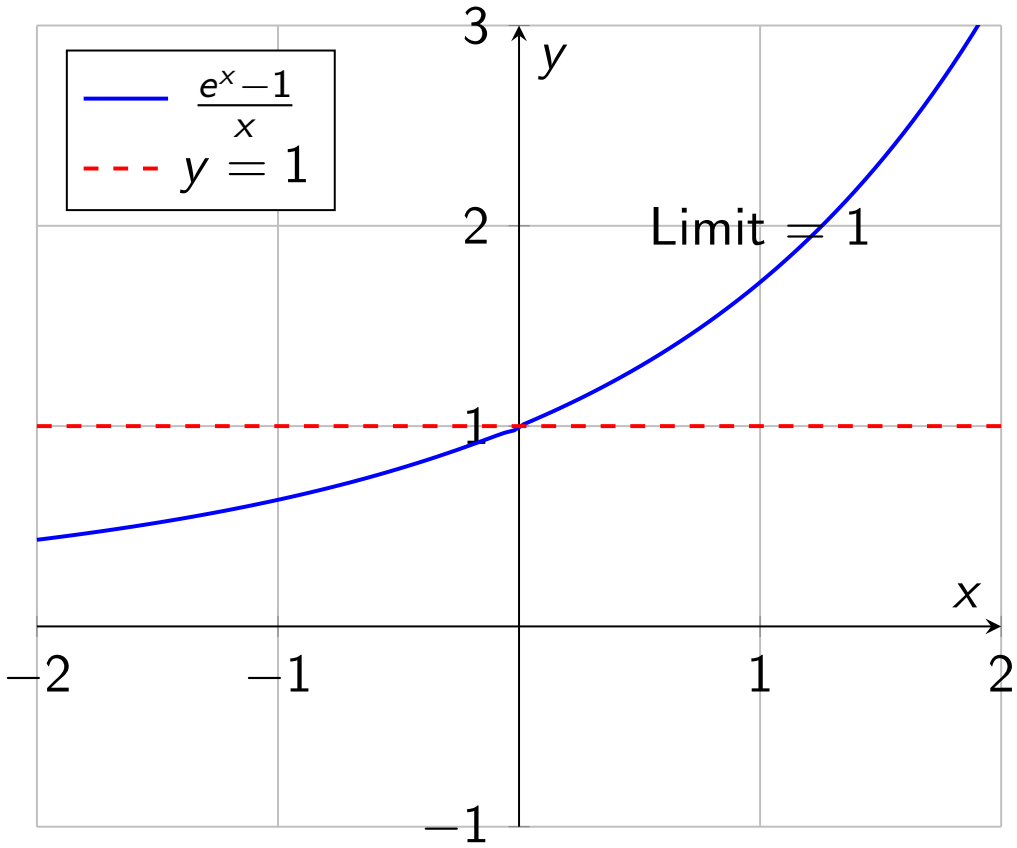
Evaluate: $\lim_{x\to 0} \frac{e^x-1}{x}$

Solution

1. Check form: As $x \to 0$, numerator $\to e^0 - 1 = 0$, denominator $\to 0$ $\to \frac{0}{0}$
2. Apply L'Hôpital's Rule:

$$\lim_{x\to 0} \frac{e^x - 1}{x} = \lim_{x\to 0} \frac{\frac{d}{dx}[e^x - 1]}{\frac{d}{dx}[x]} = \lim_{x\to 0} \frac{e^x}{1}$$

3. Evaluate: $\lim_{x\to 0} e^x = e^0 = 1$


$\frac{e^x-1}{x}$
$y = 1$
Limit $= 1$
3
2
1
-1
-2
-1
1
2
y
x

Problem 1: $\frac{0}{0}$ Form

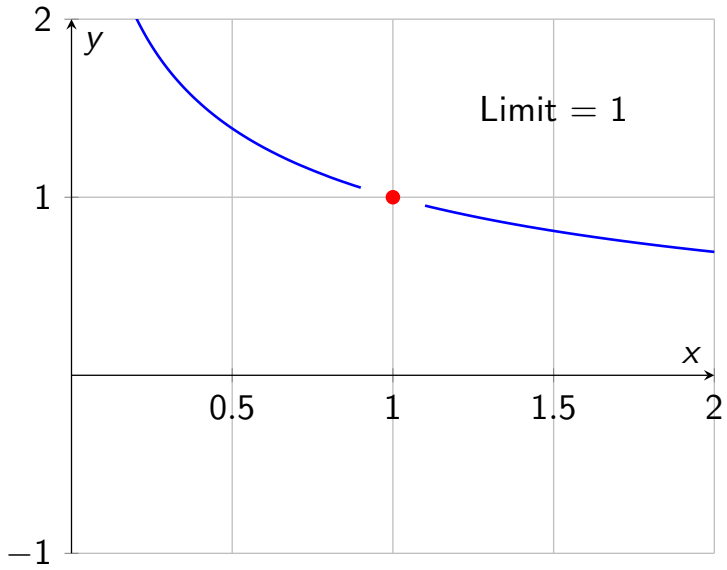
Evaluate: $\lim_{x \to 1} \frac{\ln x}{x-1}$

Solution 1: $\frac{0}{0}$ Form

1. Check form: As $x \to 1$, numerator $\to \ln 1 = 0$, denominator $\to 0 \to \frac{0}{0}$
2. Apply L'Hôpital's Rule:

$$\lim_{x \to 1} \frac{\ln x}{x - 1} = \lim_{x \to 1} \frac{\frac{d}{dx}[\ln x]}{\frac{d}{dx}[x - 1]} = \lim_{x \to 1} \frac{1/x}{1}$$

3. Evaluate: $\lim_{x \to 1} \frac{1}{x} = 1$



Type 2: $\frac{\infty}{\infty}$ Form

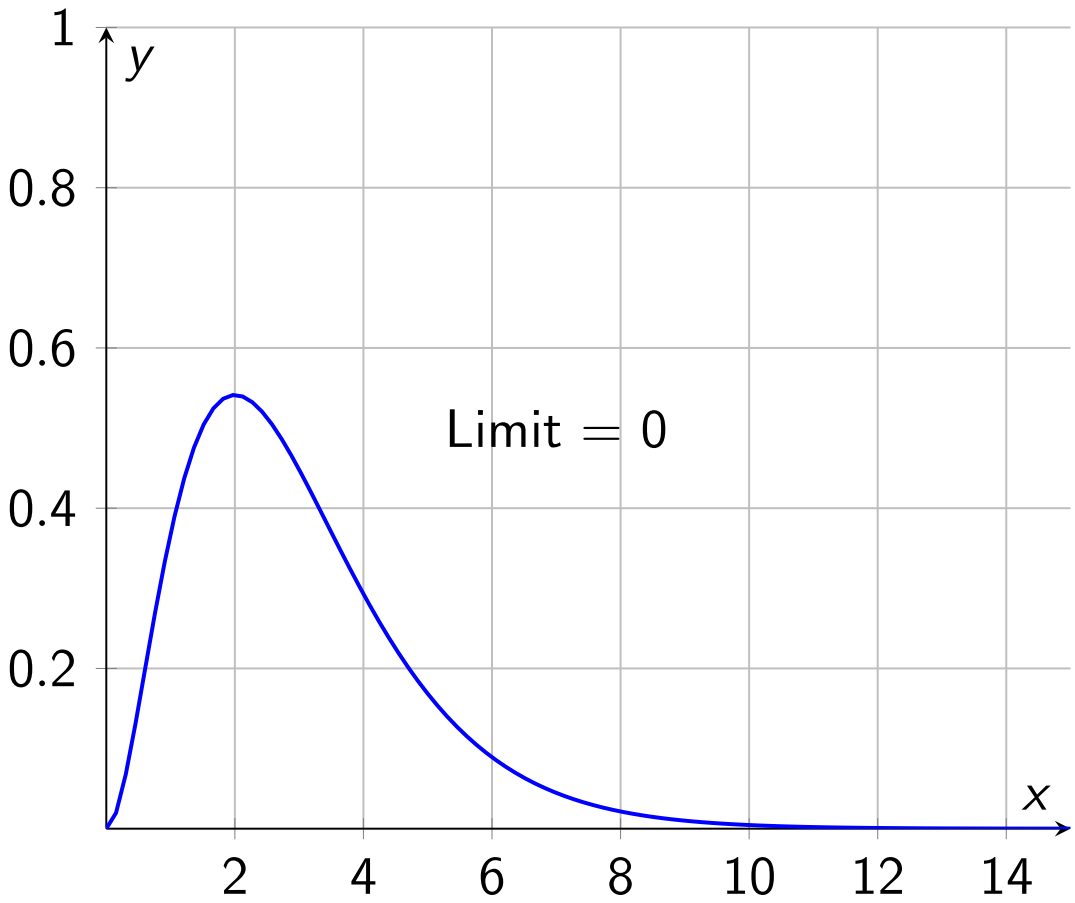
Example

Evaluate: $\lim_{x\to\infty} \frac{x^2}{e^x}$

Solution

1. Check form: As $x \to \infty$, numerator $\to \infty$, denominator $\to \infty \to \frac{\infty}{\infty}$
2. Apply L'Hôpital's Rule (may need multiple applications):

$$\lim_{x\to\infty} \frac{x^2}{e^x} = \lim_{x\to\infty} \frac{2x}{e^x} \quad (\text{still } \frac{\infty}{\infty})$$
$$= \lim_{x\to\infty} \frac{2}{e^x} = 0$$


1
y
0.8
0.6
Limit = 0
0.4
0.2
x
2
4
6
8
10
12
14

Problem 2: $\frac{\infty}{\infty}$ Form

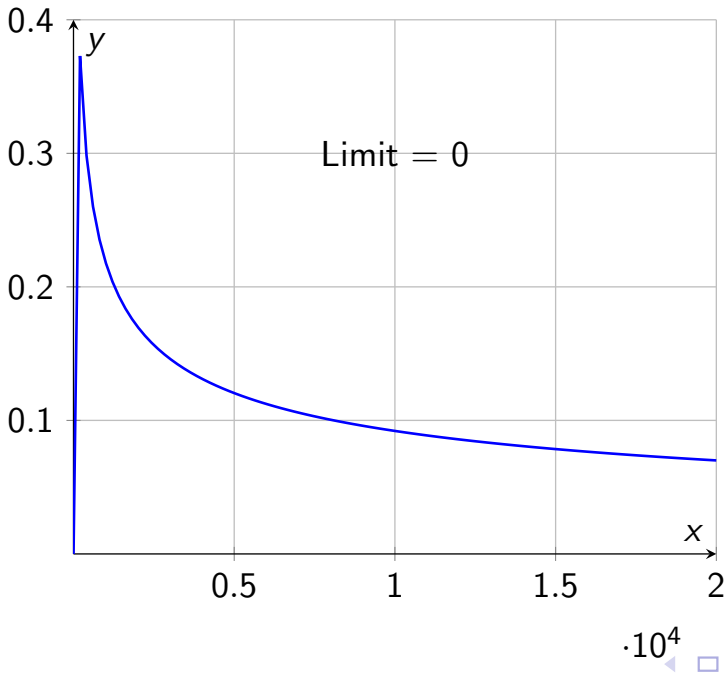
Evaluate: $\lim_{x\to\infty} \frac{\ln x}{\sqrt{x}}$

Solution 2: $\frac{\infty}{\infty}$ Form

1. Check form: As $x \to \infty$, numerator $\to \infty$, denominator $\to \infty \to \frac{\infty}{\infty}$
2. Apply L'Hôpital's Rule:

$$\lim_{x\to\infty} \frac{\ln x}{\sqrt{x}} = \lim_{x\to\infty} \frac{1/x}{\frac{1}{2\sqrt{x}}}$$

$$= \lim_{x\to\infty} \frac{1/x}{\frac{1}{2\sqrt{x}}} = \lim_{x\to\infty} \frac{2\sqrt{x}}{x} = \lim_{x\to\infty} \frac{2}{\sqrt{x}} = 0$$



Type 3: $0 \cdot \infty$ Form

Strategy

Convert to $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form by rewriting:

$$0 \cdot \infty = \frac{0}{1/\infty} = \frac{0}{0} \quad \text{or} \quad \frac{\infty}{1/0} = \frac{\infty}{\infty}$$

Example

Evaluate: $\lim_{x \to 0^+} x \ln x$

1. Rewrite as $\frac{\infty}{\infty}$ form: $\lim_{x \to 0^+} \frac{\ln x}{1/x}$
2. Apply L'Hôpital's Rule:

$$\begin{aligned} \lim_{x \to 0^+} \frac{\ln x}{1/x} &= \lim_{x \to 0^+} \frac{1/x}{-1/x^2} \\ &= \lim_{x \to 0^+} (-x) = 0 \end{aligned}$$

Problem 3: $0 \cdot \infty$ Form

Evaluate: $\lim_{x \to 0} x \cot x$

Solution 3: $0 \cdot \infty$ Form

1. Rewrite using trigonometric identity: $\cot x = \frac{\cos x}{\sin x}$

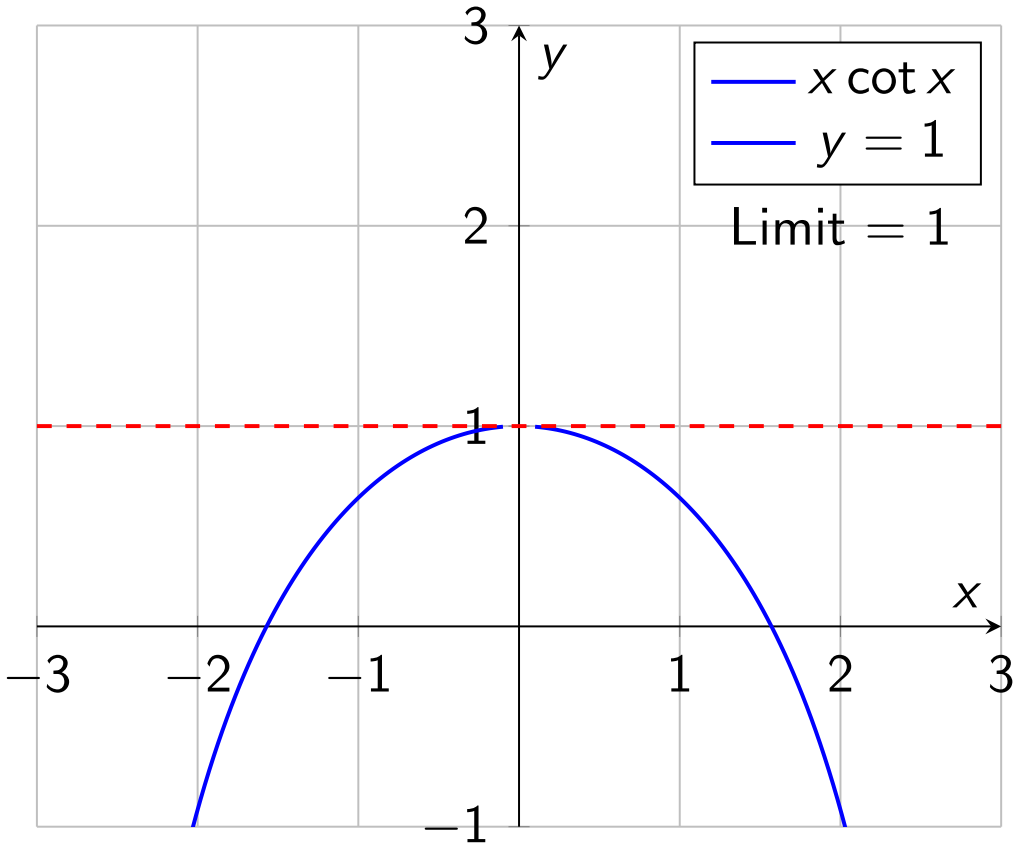
$$\lim_{x \to 0} x \cot x = \lim_{x \to 0} x \cdot \frac{\cos x}{\sin x} = \lim_{x \to 0} \frac{x \cos x}{\sin x}$$

2. Now we have $\frac{0}{0}$ form as $x \to 0$
3. Apply L'Hôpital's Rule:

$$\lim_{x \to 0} \frac{x \cos x}{\sin x} = \lim_{x \to 0} \frac{\cos x - x \sin x}{\cos x}$$

4. Evaluate:

$$\lim_{x \to 0} \frac{\cos x - x \sin x}{\cos x} = \frac{\cos 0 - 0 \cdot \sin 0}{\cos 0} = \frac{1 - 0}{1} = 1$$


3
y
2
1
−1
x
−3
−2
−1
1
2
3
x cot x
y = 1
Limit = 1

Type 4: $\infty - \infty$ Form

Strategy

Combine into a single fraction or use a common denominator to convert to $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form.

Example

Evaluate: $\lim_{x\to 0}\left(\frac{1}{x} - \frac{1}{e^x-1}\right)$

1. Common denominator: $\frac{e^x-1-x}{x(e^x-1)}$
2. Check form: $\frac{0}{0}$ as $x \to 0$
3. Apply L'Hôpital's Rule:

$$\lim_{x\to 0}\frac{e^x-1-x}{x(e^x-1)} = \lim_{x\to 0}\frac{e^x-1}{e^x-1+xe^x} \quad (\text{still } \frac{0}{0})$$

$$= \lim_{x\to 0}\frac{e^x}{e^x+e^x+xe^x} = \lim_{x\to 0}\frac{e^x}{2e^x+xe^x} = \frac{1}{2}$$

Type 5: Exponential Forms (0^0, 1^∞, ∞^0)

Strategy

Use logarithmic transformation: Let $y = f(x)^{g(x)}$, then $\ln y = g(x) \ln f(x)$

Example

Evaluate: $\lim_{x \to 0^+} x^x$

Solution

1. Let $y = x^x$, then $\ln y = x \ln x$
2. $\lim_{x \to 0^+} \ln y = \lim_{x \to 0^+} x \ln x = 0$ (from previous example)
   item So $\lim_{x \to 0^+} y = e^0 = 1$

Problem 4: 1^{∞} Form

Evaluate: $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{x}$

Solution 4: 1^∞ Form

1. Let $y = \left(1 + \frac{1}{x}\right)^x$, then $\ln y = x \ln\left(1 + \frac{1}{x}\right)$
2. $\lim_{x\to\infty} \ln y = \lim_{x\to\infty} x \ln\left(1 + \frac{1}{x}\right)$
3. Rewrite as $\frac{\ln(1+1/x)}{1/x}$ (form $\frac{0}{0}$)
4. Apply L'Hôpital's Rule:

$$\lim_{x\to\infty} \frac{\ln(1+1/x)}{1/x} = \lim_{x\to\infty} \frac{\frac{-1/x^2}{1+1/x}}{-1/x^2}$$
$$= \lim_{x\to\infty} \frac{1}{1+1/x} = 1$$

5. So $\lim_{x\to\infty} y = e^1 = e$

Famous Result

This proves that $\lim_{x\to\infty} \left(1 + \frac{1}{x}\right)^x = e$

Multiple Applications of L'Hôpital's Rule

Example

Evaluate: $\lim_{x\to 0} \frac{x-\sin x}{x^3}$

Solution

$$\begin{aligned}\lim_{x\to 0} \frac{x-\sin x}{x^3} &= \lim_{x\to 0} \frac{1-\cos x}{3x^2} \quad (\text{still } \frac{0}{0}) \\ &= \lim_{x\to 0} \frac{\sin x}{6x} \quad (\text{still } \frac{0}{0}) \\ &= \lim_{x\to 0} \frac{\cos x}{6} = \frac{1}{6}\end{aligned}$$

Keep Applying!

Apply L'Hôpital's Rule repeatedly until you no longer get an indeterminate form.

When L'Hôpital's Rule Fails

Example

Evaluate: $\lim_{x\to\infty} \frac{x+\sin x}{x}$

Incorrect Application

$$\lim_{x\to\infty} \frac{x+\sin x}{x} = \lim_{x\to\infty} \frac{1+\cos x}{1} \quad \text{Does not exist!}$$

Correct Solution

$$\lim_{x\to\infty} \frac{x+\sin x}{x} = \lim_{x\to\infty} \left(1+\frac{\sin x}{x}\right) = 1+0=1$$

Warning

L'Hôpital's Rule may fail if the limit of derivatives doesn't exist. Always check if simpler methods work!

Advanced Example: Trigonometric Limits

Example

Evaluate: $\lim_{x\to 0} \frac{\tan x - x}{x^3}$

Solution

$$\lim_{x\to 0} \frac{\tan x - x}{x^3} = \lim_{x\to 0} \frac{\sec^2 x - 1}{3x^2}$$

$$= \lim_{x\to 0} \frac{\tan^2 x}{3x^2} = \frac{1}{3} \lim_{x\to 0} \left(\frac{\tan x}{x}\right)^2$$

$$= \frac{1}{3}(1)^2 = \frac{1}{3}$$

Practice Problems

1. $\lim_{x\to 0} \frac{\sin(3x)}{2x}$
2. $\lim_{x\to 1} \frac{x^3-1}{x^2-1}$
3. $\lim_{x\to \infty} \frac{x^3}{e^{2x}}$
4. $\lim_{x\to 0^+} x^2 \ln x$
5. $\lim_{x\to \frac{\pi}{2}} (\sec x - \tan x)$
6. $\lim_{x\to 0} (1+2x)^{1/x}$
7. $\lim_{x\to 0} \frac{e^x - e^{-x} - 2x}{x^3}$
8. $\lim_{x\to \infty} \frac{\sqrt{x}}{\ln x}$
9. $\lim_{x\to 0} \frac{\arcsin x}{x}$
10. $\lim_{x\to 1} \frac{x^x - x}{\ln x - x + 1}$

Practice Problem Solutions

1. $\frac{3}{2}$
2. $\frac{3}{2}$
3. 0
4. 0
5. 0
6. e^2
7. $\frac{1}{3}$
8. ∞
9. 1
10. 2

Summary: L'Hôpital's Rule Strategy

Step-by-Step Approach

1. **Check form**: Must be $\frac{0}{0}$ or $\frac{\infty}{\infty}$
2. **Simplify**: Try algebraic manipulation first
3. **Apply Rule**: Differentiate numerator and denominator separately
4. **Repeat**: If still indeterminate, apply rule again
5. **Verify**: Check if limit exists

Special Forms Conversion

- $0 \cdot \infty$: Rewrite as $\frac{0}{1/\infty}$ or $\frac{\infty}{1/0}$
- $\infty - \infty$: Combine into single fraction
- $0^0, 1^\infty, \infty^0$: Use logarithmic transformation

Common Mistakes to Avoid

Don't Make These Errors!

- Applying to forms that aren't $\frac{0}{0}$ or $\frac{\infty}{\infty}$
- Applying when simpler algebraic methods work
- Applying when limit of derivatives doesn't exist
- Using quotient rule instead of differentiating separately
- Forgetting to check conditions

Verification Checklist

- Are both numerator and denominator approaching 0 or ∞?
- Are the functions differentiable near the point?
- Is $g'(x) \neq 0$ near the point?
- Does the limit of the ratio of derivatives exist?

Advanced Applications

L'Hôpital's Rule helps determine which functions grow faster:

- $\lim_{x\to\infty} \frac{\ln x}{x^p} = 0$ for $p > 0$
- $\lim_{x\to\infty} \frac{x^p}{e^x} = 0$ for all p
- $\lim_{x\to\infty} \frac{e^x}{x!} = 0$

Real-World Applications

- Physics: Rates of change in motion
- Economics: Marginal analysis
- Engineering: Signal processing
- Computer Science: Algorithm complexity analysis

Remember

L'Hôpital's Rule is a powerful tool, but always think about whether it's the right tool for the job!